

TURBOFAN ENGINE FAILURE PREDICTION USING RUL

PROBLEM STATEMENT

Aircraft turbofan engines gradually degrade due to continuous operational stress, temperature variations, and mechanical wear. Failure to detect this degradation early can lead to unexpected engine breakdowns, safety risks, and high maintenance costs.

However, traditional maintenance methods rely on fixed schedules rather than the actual condition of the engine, making them inefficient and reactive.

DATASET OVERVIEW

- ▶ The **CMAPSS dataset** from Kaggle is divided into **train and test sets**, along with a separate **RUL (Remaining Useful Life) file**.
- ▶ **Train dataset:** Contains time-series sensor and operational data for multiple engines **from start until failure**, used to train the model.
- ▶ **Test dataset:** Contains sensor and operational data for engines **up to a point before failure**, used to evaluate model predictions.
- ▶ **RUL dataset:** Provides the **remaining useful life for each engine in the test set**, which serves as the ground truth for model validation.

[4]:

	0	1	2	3	4	5	6	7	8	9	...	18	19	20	21	22	23	24	25	26	27
0	1	1	-0.0007	-0.0004	100.0	518.67	641.82	1589.70	1400.60	14.62	...	8138.62	8.4195	0.03	392	2388	100.0	39.06	23.4190	NaN	NaN
1	1	2	0.0019	-0.0003	100.0	518.67	642.15	1591.82	1403.14	14.62	...	8131.49	8.4318	0.03	392	2388	100.0	39.00	23.4236	NaN	NaN
2	1	3	-0.0043	0.0003	100.0	518.67	642.35	1587.99	1404.20	14.62	...	8133.23	8.4178	0.03	390	2388	100.0	38.95	23.3442	NaN	NaN
3	1	4	0.0007	0.0000	100.0	518.67	642.35	1582.79	1401.87	14.62	...	8133.83	8.3682	0.03	392	2388	100.0	38.88	23.3739	NaN	NaN
4	1	5	-0.0019	-0.0002	100.0	518.67	642.37	1582.85	1406.22	14.62	...	8133.80	8.4294	0.03	393	2388	100.0	38.90	23.4044	NaN	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
61244	249	251	9.9998	0.2500	100.0	489.05	605.33	1516.36	1315.28	10.52	...	8185.69	8.4541	0.03	372	2319	100.0	29.11	17.5234	NaN	NaN
61245	249	252	0.0028	0.0015	100.0	518.67	643.42	1598.92	1426.77	14.62	...	8185.47	8.2221	0.03	396	2388	100.0	39.38	23.7151	NaN	NaN
61246	249	253	0.0029	0.0000	100.0	518.67	643.68	1607.72	1430.56	14.62	...	8193.94	8.2525	0.03	395	2388	100.0	39.78	23.8270	NaN	NaN
61247	249	254	35.0046	0.8400	100.0	449.44	555.77	1381.29	1148.18	5.48	...	8125.64	9.0515	0.02	337	2223	100.0	15.26	9.0774	NaN	NaN
61248	249	255	42.0030	0.8400	100.0	445.00	549.85	1369.75	1147.45	3.91	...	8144.33	9.1207	0.02	333	2212	100.0	10.66	6.4341	NaN	NaN

160359 rows × 28 columns

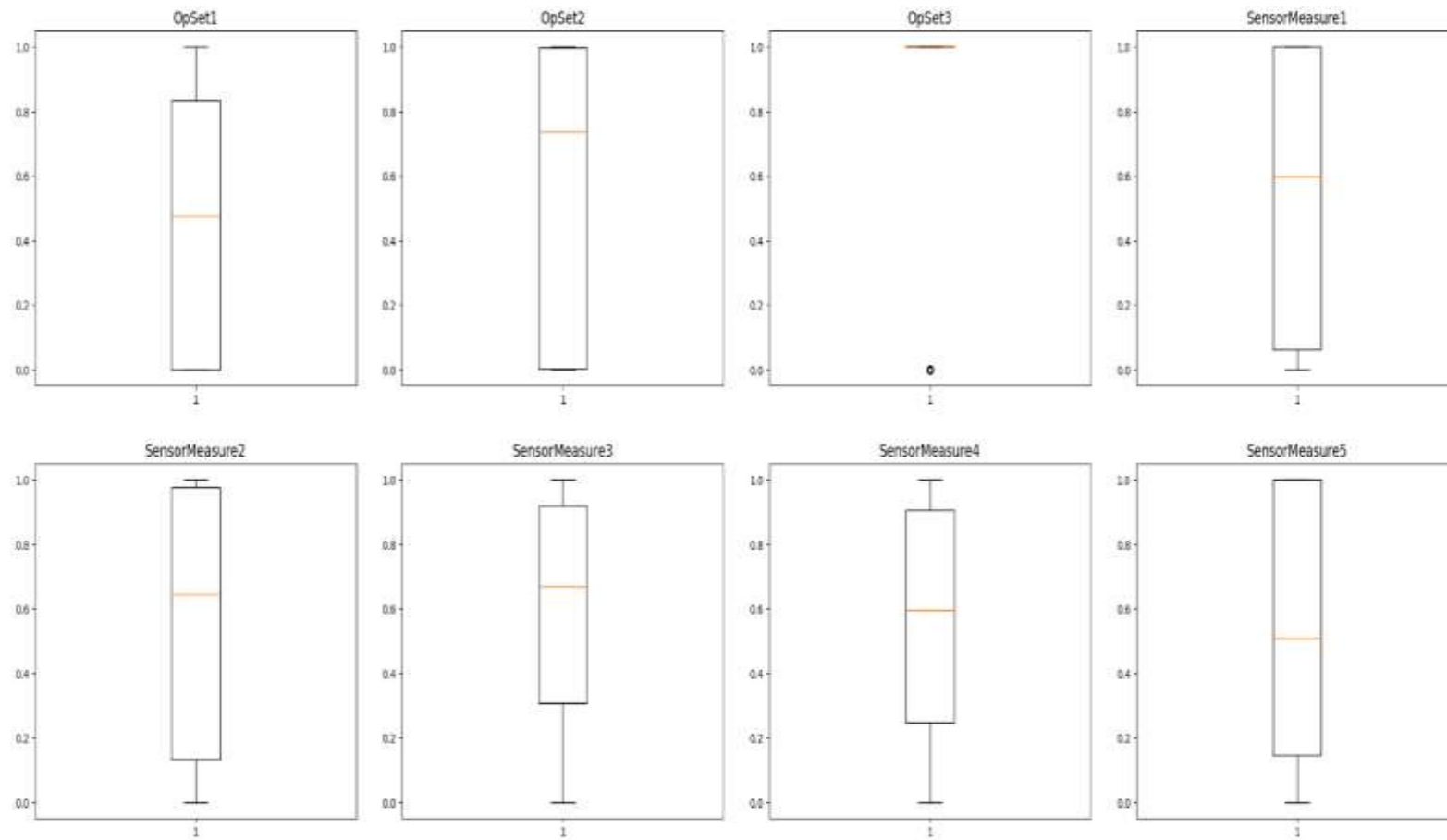
FEATURES & TARGET

- ❖ The dataset contains **26 columns/features**:
- ❖ **1 Engine ID** - Identifies the engine.
- ❖ **1 Time Cycle** - Records the time step or operational cycle.
- ❖ **3 Operational Settings** - Variables such as altitude, Mach number, throttle, etc., that describe engine operating conditions.
- ❖ **21 Sensor Measurements** - Sensors monitoring engine performance, e.g., temperature, pressure, fan speed, vibration, etc.
- ❖ **Output / Target Value:**
- ❖ The main output used for prediction is the **Remaining Useful Life (RUL)** of each engine, which indicates how many cycles remain before failure.
- ❖ **Important Features:**
- ❖ Operational settings (OS1-OS3) help account for varying conditions.
- ❖ Critical sensor readings such as **temperature, pressure, and fan speed** are often the most predictive for RUL estimation.

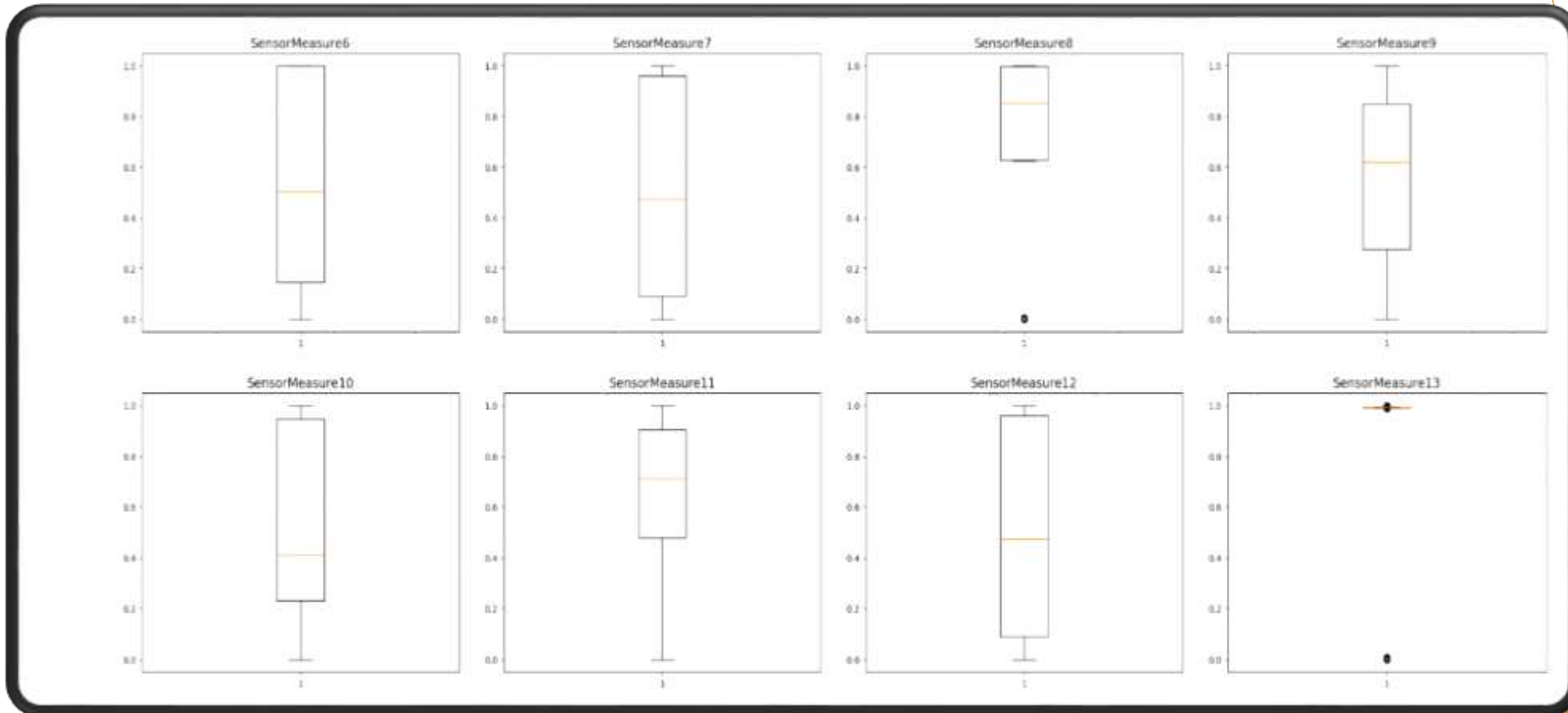
FEATURES WITH TARGET VARIABLE

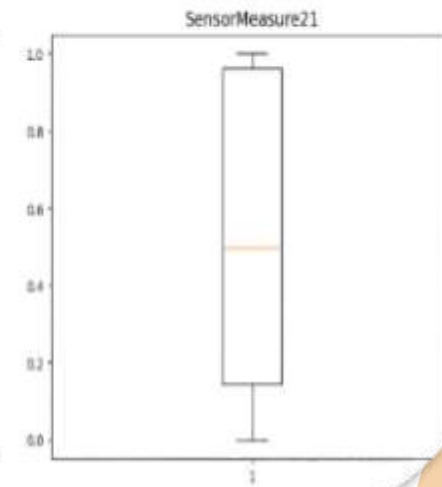
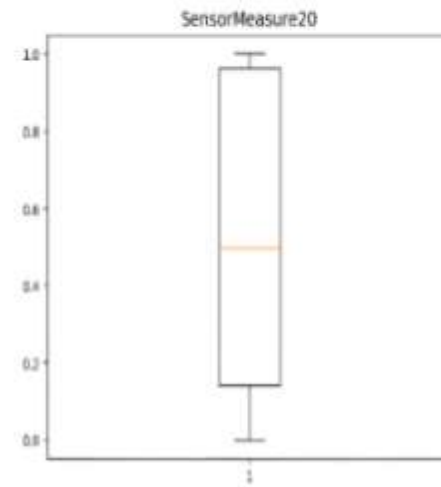
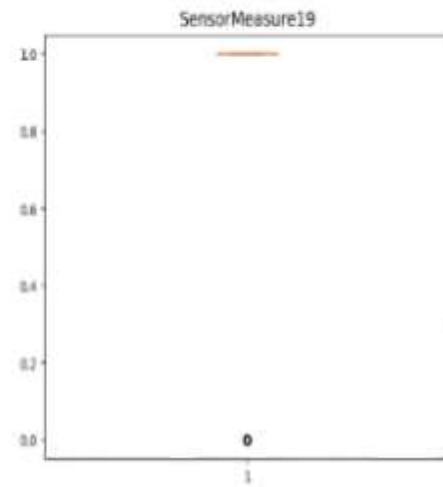
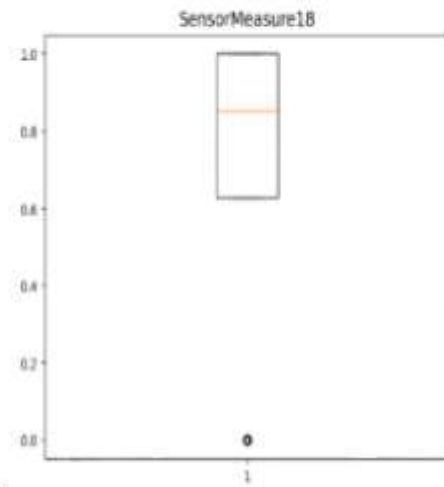
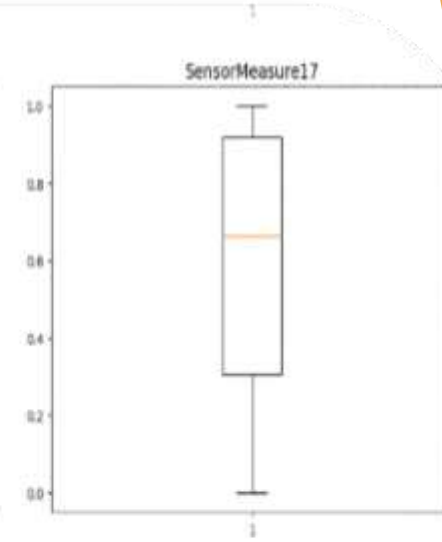
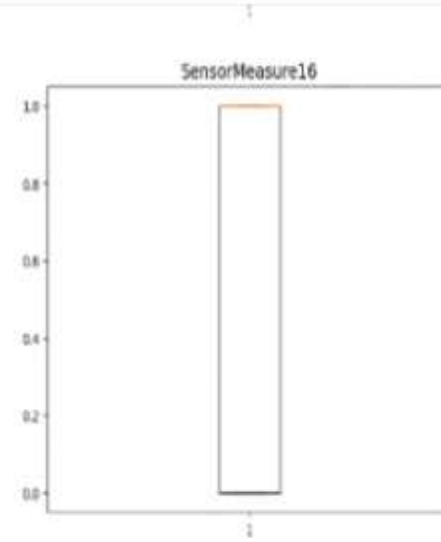
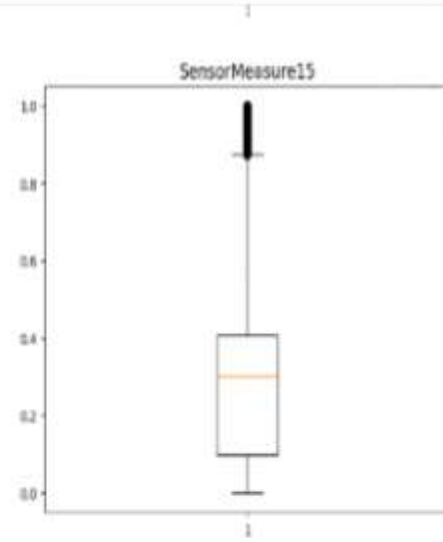
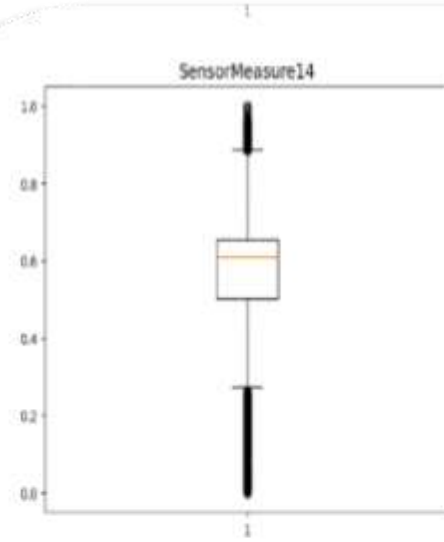
```
ID  
Cycle  
OpSet1  
OpSet2  
OpSet3  
SensorMeasure1  
SensorMeasure2  
SensorMeasure3  
SensorMeasure4  
SensorMeasure5  
SensorMeasure6  
SensorMeasure7  
SensorMeasure8  
SensorMeasure9  
SensorMeasure10  
SensorMeasure11  
SensorMeasure12  
SensorMeasure13  
SensorMeasure14  
SensorMeasure15  
SensorMeasure16  
SensorMeasure17  
SensorMeasure18  
SensorMeasure19  
SensorMeasure20  
SensorMeasure21  
RUL
```

EDA

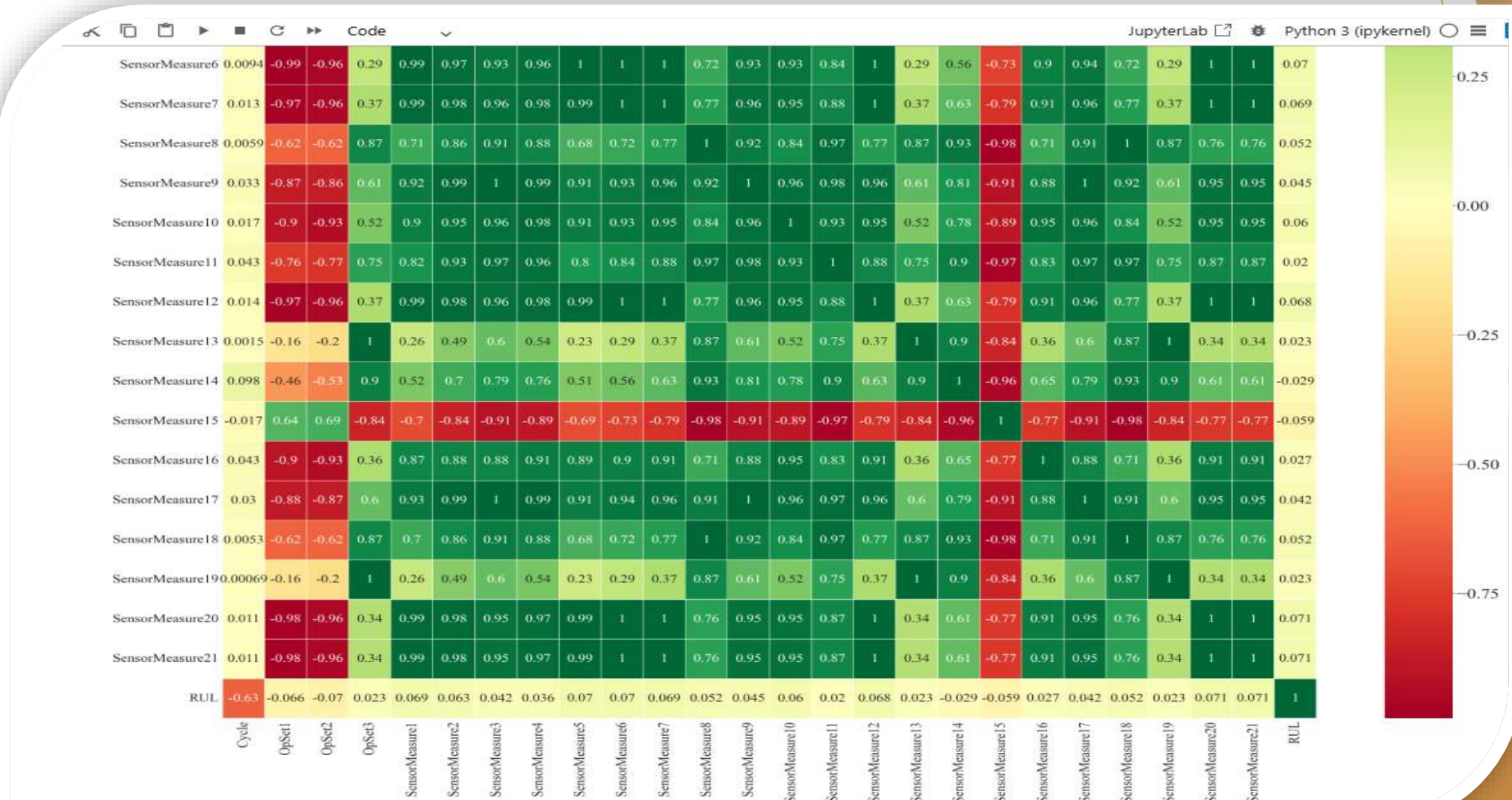


Box plots of all Features

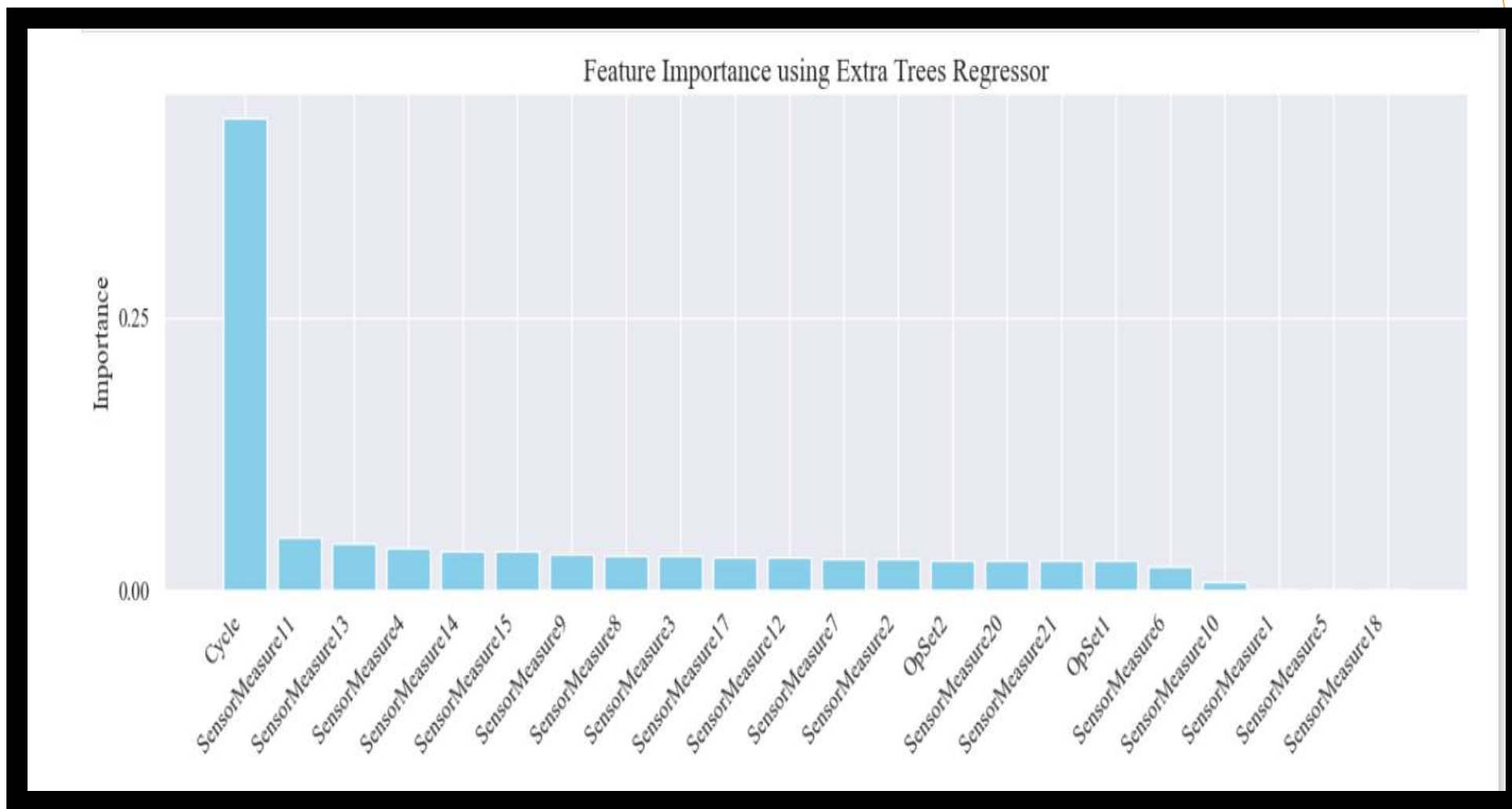




CORRELATION HEATMAP



FEATURE IMPORTANCE



Finding Unique Numeric values in each Feature

```
[4]: df.nunique(axis=0)
```

```
[4]: Cycle          543  
OpSet1          670  
OpSet2          111  
OpSet3           2  
SensorMeasure1    6  
SensorMeasure2   1799  
SensorMeasure3  15377  
SensorMeasure4  20582  
SensorMeasure5     6  
SensorMeasure6    56  
SensorMeasure7   6311  
SensorMeasure8   1163  
SensorMeasure9  34161  
SensorMeasure10   21  
SensorMeasure11   771  
SensorMeasure12  5982  
SensorMeasure13   567  
SensorMeasure14  20090  
SensorMeasure15  13124  
SensorMeasure16     2  
SensorMeasure17    58  
SensorMeasure18     6  
SensorMeasure19     2  
SensorMeasure20   676  
SensorMeasure21  26659  
RUL             543  
dtype: int64
```

FEATURE SELECTION

```
Features Used: 1  
Selected Columns: ['Cycle']  
Train RMSE: 73.5074  
Test RMSE: 68.1698
```

```
Features Used: 2  
Selected Columns: ['Cycle', 'SensorMeasure11']  
Train RMSE: 51.2333  
Test RMSE: 72.8393
```

```
Features Used: 3  
Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13']  
Train RMSE: 30.2944  
Test RMSE: 68.6452
```

```
Features Used: 4  
Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4']  
Train RMSE: 26.6424  
Test RMSE: 66.1088
```

```
Features Used: 5  
Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4', 'SensorMeasure14']  
Train RMSE: 26.0769  
Test RMSE: 65.5132
```

```
Features Used: 6  
Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4', 'SensorMeasure14', 'SensorMeasure15']  
Train RMSE: 25.8624  
Test RMSE: 64.5865
```

```
Features Used: 7  
Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4', 'SensorMeasure14', 'SensorMeasure15', 'SensorMeasure9']  
Train RMSE: 25.7295  
Test RMSE: 64.4539
```

Features Used: 18

Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4', 'SensorMeasure14', 'SensorMeasure15', 'SensorMeasure9', 'SensorMeasure8', 'SensorMeasure3', 'SensorMeasure17', 'SensorMeasure12', 'SensorMeasure7', 'SensorMeasure2', 'OpSet2', 'SensorMeasure20', 'SensorMeasure21', 'OpSet1', 'SensorMeasure6']

Train RMSE: 25.3231

Test RMSE: 63.1403

Features Used: 19

Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4', 'SensorMeasure14', 'SensorMeasure15', 'SensorMeasure9', 'SensorMeasure8', 'SensorMeasure3', 'SensorMeasure17', 'SensorMeasure12', 'SensorMeasure7', 'SensorMeasure2', 'OpSet2', 'SensorMeasure20', 'SensorMeasure21', 'OpSet1', 'SensorMeasure6', 'SensorMeasure10']

Train RMSE: 25.3168

Test RMSE: 63.1550

Features Used: 20

Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4', 'SensorMeasure14', 'SensorMeasure15', 'SensorMeasure9', 'SensorMeasure8', 'SensorMeasure3', 'SensorMeasure17', 'SensorMeasure12', 'SensorMeasure7', 'SensorMeasure2', 'OpSet2', 'SensorMeasure20', 'SensorMeasure21', 'OpSet1', 'SensorMeasure6', 'SensorMeasure10', 'SensorMeasure1']

Train RMSE: 25.3152

Test RMSE: 63.1419

Features Used: 21

Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4', 'SensorMeasure14', 'SensorMeasure15', 'SensorMeasure9', 'SensorMeasure8', 'SensorMeasure3', 'SensorMeasure17', 'SensorMeasure12', 'SensorMeasure7', 'SensorMeasure2', 'OpSet2', 'SensorMeasure20', 'SensorMeasure21', 'OpSet1', 'SensorMeasure6', 'SensorMeasure10', 'SensorMeasure1', 'SensorMeasure5']

Train RMSE: 25.3316

Test RMSE: 63.1264

Features Used: 22

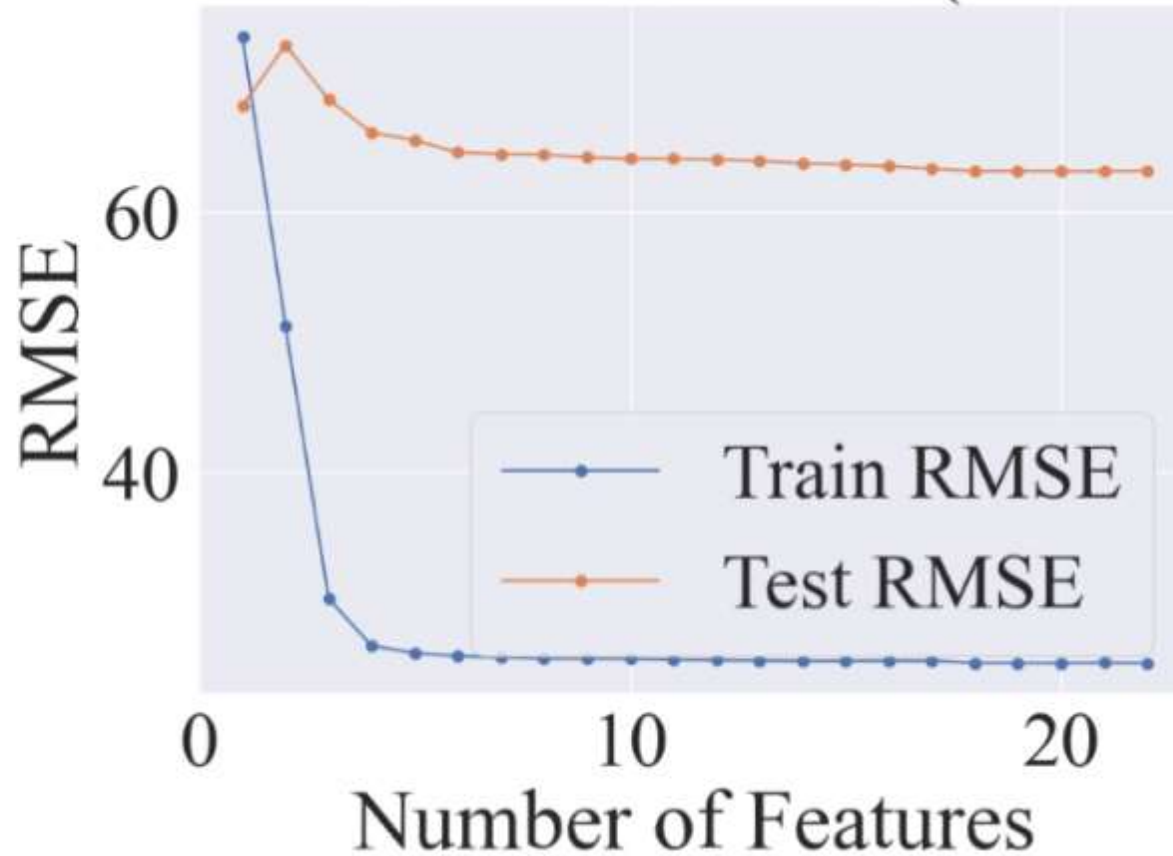
Selected Columns: ['Cycle', 'SensorMeasure11', 'SensorMeasure13', 'SensorMeasure4', 'SensorMeasure14', 'SensorMeasure15', 'SensorMeasure9', 'SensorMeasure8', 'SensorMeasure3', 'SensorMeasure17', 'SensorMeasure12', 'SensorMeasure7', 'SensorMeasure2', 'OpSet2', 'SensorMeasure20', 'SensorMeasure21', 'OpSet1', 'SensorMeasure6', 'SensorMeasure10', 'SensorMeasure1', 'SensorMeasure5', 'SensorMeasure18']

Train RMSE: 25.3070

Test RMSE: 63.1621

SCATTER PLOT

RMSE vs Number of Features (Random Forest)



Dataset Before Model Training

Delimiter: , v							
	Cycle	SensorMeasure11	SensorMeasure13	SensorMeasure4	SensorMeasure14	SensorMeasure15	SensorMeasure9
1	1	47.47	2388.02	1400.6	8138.62	8.4195	9046.19
2	2	47.49	2388.07	1403.14	8131.49	8.4318	9044.07
3	3	47.27	2388.03	1404.2	8133.23	8.4178	9052.94
4	4	47.13	2388.08	1401.87	8133.83	8.3682	9049.48
5	5	47.28	2388.04	1406.22	8133.8	8.4294	9055.15
6	6	47.16	2388.03	1398.37	8132.85	8.4108	9049.68
7	7	47.36	2388.03	1397.77	8132.32	8.3974	9059.13
8	8	47.24	2388.03	1400.97	8131.07	8.4076	9040.8
9	9	47.29	2388.05	1394.8	8125.69	8.3728	9046.46
10	10	47.03	2388.06	1400.46	8129.38	8.4286	9051.7
11	11	47.15	2388.01	1400.64	8140.58	8.434	9049.61
12	12	47.18	2388.02	1400.15	8134.25	8.3938	9049.37
13	13	47.38	2388.08	1400.83	8128.1	8.4152	9046.82
14	14	47.44	2388.0	1399.16	8134.43	8.3964	9047.37
15	15	47.3	2388.08	1402.13	8127.56	8.4199	9052.22
16	16	47.24	2388.07	1404.5	8136.11	8.3936	9049.34
17	17	47.12	2388.04	1399.95	8137.27	8.4542	9054.92
18	18	47.21	2388.07	1396.12	8132.73	8.4028	9049.55
19	19	47.4	2388.03	1400.35	8129.13	8.4321	9053.99
20	20	47.22	2388.02	1405.23	8129.71	8.421	9045.9
21	21	47.15	2388.08	1398.13	8134.02	8.4049	9048.15
22	22	47.24	2388.03	1400.57	8130.41	8.4034	9061.21
23	23	47.25	2388.05	1394.75	8127.9	8.424	9046.28

Delimiter: ,

	SensorMeasure21	OpSet1	SensorMeasure6	SensorMeasure10	SensorMeasure1	SensorMeasure5	SensorMeasure18	RUL
1	23.419	-0.0007	21.61	1.3	518.67	14.62	2388	320
2	23.4236	0.0019	21.61	1.3	518.67	14.62	2388	319
3	23.3442	-0.0043	21.61	1.3	518.67	14.62	2388	318
4	23.3739	0.0007	21.61	1.3	518.67	14.62	2388	317
5	23.4044	-0.0019	21.61	1.3	518.67	14.62	2388	316
6	23.3669	-0.0043	21.61	1.3	518.67	14.62	2388	315
7	23.3774	0.001	21.61	1.3	518.67	14.62	2388	314
8	23.3106	-0.0034	21.61	1.3	518.67	14.62	2388	313
9	23.4066	0.0008	21.61	1.3	518.67	14.62	2388	312
10	23.4694	-0.0033	21.61	1.3	518.67	14.62	2388	311
11	23.4787	0.0018	21.61	1.3	518.67	14.62	2388	310
12	23.366	0.0016	21.61	1.3	518.67	14.62	2388	309
13	23.2757	-0.0019	21.61	1.3	518.67	14.62	2388	308
14	23.3826	0.0009	21.61	1.3	518.67	14.62	2388	307
15	23.35	-0.0018	21.61	1.3	518.67	14.62	2388	306
16	23.455	0.0006	21.61	1.3	518.67	14.62	2388	305
17	23.3319	0.0002	21.61	1.3	518.67	14.62	2388	304
18	23.3987	-0.0031	21.61	1.3	518.67	14.62	2388	303
19	23.3464	0.0032	21.61	1.3	518.67	14.62	2388	302
20	23.422	-0.0037	21.61	1.3	518.67	14.62	2388	301
21	23.3101	-0.0012	21.61	1.3	518.67	14.62	2388	300
22	23.3792	0.0002	21.61	1.3	518.67	14.62	2388	299
23	23.4562	0.0034	21.61	1.3	518.67	14.62	2388	298

ML MODELS USED

- **Linear Regression (LR):** A simple baseline model that fits a straight-line relationship to predict the RUL based on sensor features.
- **Ridge Regression (RR):** An improved linear model that reduces overfitting by applying L2 regularization to handle multicollinearity in sensor data.
- **Random Forest:** An ensemble of multiple decision trees that captures complex nonlinear patterns in engine degradation for better RUL prediction.
- **Gradient Boosting:** A powerful boosting algorithm that builds sequential trees to minimize prediction error and extract subtle degradation trends.
- **XGBoost:** An optimized gradient boosting model that uses regularization and efficient computation to achieve highly accurate and robust RUL predictions.

METRICS USED

- **RMSE (Root Mean Squared Error):** Measures how far the model's predictions are from actual values, giving higher weight to larger errors.
- **MAE (Mean Absolute Error):** Calculates the average absolute difference between predicted and actual values, treating all errors equally.
- **R^2 Score (Coefficient of Determination):** Indicates how well the model explains the variance in the target value, with values closer to 1 showing better performance

TABULATION OF METRICS of EACH MODEL AFTER TESTING

[39]:

	Model	RMSE	MAE	R2 Score
0	Linear Regression	63.0228	53.6863	0.4298
1	Ridge Regression	63.0210	53.6851	0.4298
2	Random Forest	60.0926	51.2736	0.4816
3	Gradient Boosting	59.8619	51.3850	0.4856
4	XGBoost	59.6524	51.2243	0.4891

BEST PERFORMING MODEL

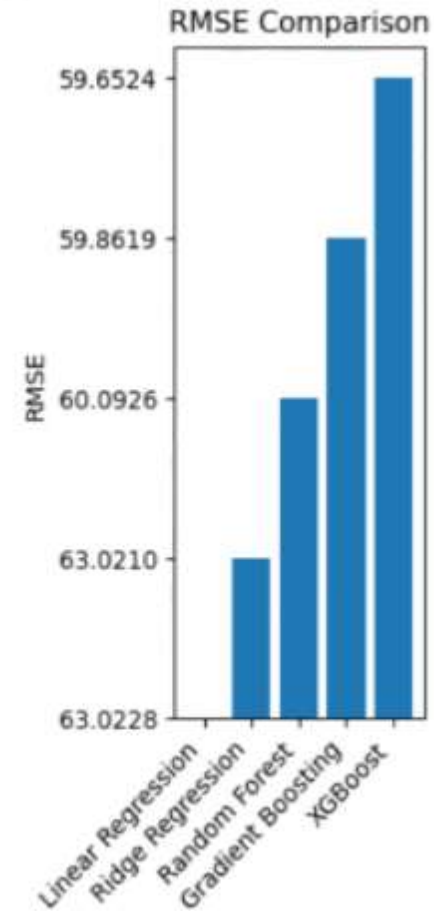
=== BEST MODEL (Based on Lowest RMSE) ===

Model: XGBoost

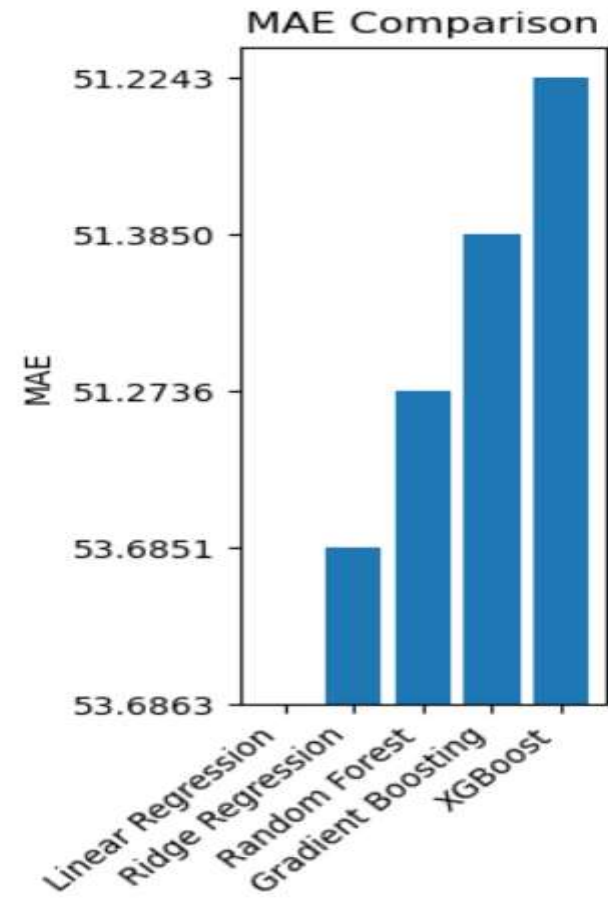
RMSE: 59.6524

MAE: 51.2243

R^2 Score: 0.4891



MAE COMPARISON



CONCLUSION

The machine learning models successfully predicted the Remaining Useful Life (RUL) of turbofan engines, with Gradient Boosting emerging as the most accurate. This project demonstrates that data-driven predictive maintenance can significantly reduce unexpected engine failures and improve operational safety.